Code No: 117EE **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, December - 2019
LINUX PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is the role of kernel in Linux OS? [2]
b) Define the basic characteristics of Linux Operating System. [3]
c) State the importance of dot/(.) and dot dot/(..) directories in the file system. [2]
d) How does file locking differ from record locking? [3]
e) Define a process and a thread. [2]
f) Differentiate between fork( ) and vfork( ) system calls. [3]
g) Why is it required to have a message queue? [2]
h) Write down the advantages and disadvantages of fifos. [3]
i) What are stream sockets and datagram sockets? [2]
j) List out the differences between IPv4 and IPv6. [3]

PART-B

(50 Marks)

2. List and explain various backup utilities and text processing utilities in Linux. [10]

OR

3.a) Write an awk script to process the marks obtained by all students of a course.
b) Briefly explain various shell variables and shell meta characters available in Linux.[5+5]

4. Write the syntax of mkdir( ), opendir( ), readdir( ), closedir( ) system calls and illustrate the same with a sample program of listing the contents of a directory. [10]

OR

5.a) What is the need of lseek( ) system call? Explain briefly.
b) Illustrate symlink( ) and link( ) system calls with an example program. [5+5]

6. What are the conditions under which a process is going to terminate under normal and abnormal conditions? How to handle the terminated process and retrieve its status? Explain. [10]

OR

7.a) How to handle a signal in Linux environment?
b) Explain kill( ) and raise( ) system calls implementation using a sample program. [5+5]

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8.a) Discuss the effect of O-NDELAY flag on pipes and fifos.

b) What happens if the cmdstring executed by popen( ) with a type of "r" write to its standard error? Write a program to illustrate popen( ). [5+5]

OR

9. Describe in detail about the implementation of process synchronization using semaphores of System V IPC mechanism. [10]

10. What is shared memory? What is the importance of it? Explain in detail about the process of "Attaching and detaching a shared memory segment" with an example. [10]

OR

11. Explain the four socket functions that pass a socket address structure from the process to the kernel. [10]

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